

LECTURE 13 : POWER QUERY - ADVANCED CLEANING & TRANSFORMATIONS

 **Today's Goal**

Take your Power Query skills to the next level with advanced cleaning and transformation techniques!

★ After Lecture 12, we continue with the powerful transformation operations that make your data clean, consistent and analysis-ready.

What we will cover today

- ✓ Split Column
- ✓ Merge Columns
- ✓ Fill Down
- ✓ Fill Up
- ✓ Practical Use Cases
- ✓ Interview Questions

1. SPLIT COLUMN

- Splits one column into multiple columns based on a delimiter, position or number of characters.
- Useful when a single column contains combined information.

How to use :

Select Column → Transform Tab → Split Column → Choose method

Use Cases



- ★ Split Full Name into First & Last Name.
- ★ Split City-State like "Lucknow-UP".
- ★ Split Product Code into Category & ID.
- ★ Split DateTime into Date and Time.

SPLIT COLUMN - METHODS								
→ By Delimiter	Split using a specific delimiter like comma, space, dash, slash etc.	Example : <table border="1"> <thead> <tr> <th>Name</th><th>First</th><th>Last</th></tr> </thead> <tbody> <tr> <td>Shivansh Yadav</td><td>Shivansh</td><td>Yadav</td></tr> </tbody> </table>	Name	First	Last	Shivansh Yadav	Shivansh	Yadav
Name	First	Last						
Shivansh Yadav	Shivansh	Yadav						
→ By Position	Split at a specific position or after n characters.	Example : <table border="1"> <thead> <tr> <th>ProductCode</th><th>Cat</th><th>SubCat</th></tr> </thead> <tbody> <tr> <td>ELEC-001-TV</td><td>ELEC</td><td>001-TV</td></tr> </tbody> </table>	ProductCode	Cat	SubCat	ELEC-001-TV	ELEC	001-TV
ProductCode	Cat	SubCat						
ELEC-001-TV	ELEC	001-TV						
→ By Number of Characters	Split the column into parts based on number of characters.	Example : <table border="1"> <thead> <tr> <th>MobileNo</th><th>Part 1</th><th>Part 2</th></tr> </thead> <tbody> <tr> <td>9876543210</td><td>987654</td><td>3210</td></tr> </tbody> </table>	MobileNo	Part 1	Part 2	9876543210	987654	3210
MobileNo	Part 1	Part 2						
9876543210	987654	3210						
→ By Lowercase to Uppercase	Split where lowercase changes to uppercase.	Example : <table border="1"> <thead> <tr> <th>ItemName</th><th>Item</th><th>Code</th></tr> </thead> <tbody> <tr> <td>PenBlue123</td><td>PenBlue</td><td>123</td></tr> </tbody> </table>	ItemName	Item	Code	PenBlue123	PenBlue	123
ItemName	Item	Code						
PenBlue123	PenBlue	123						
→ By Uppercase to Lowercase	Split where uppercase changes to lowercase.	Example : <table border="1"> <thead> <tr> <th>ItemName</th><th>Item</th><th>Code</th></tr> </thead> <tbody> <tr> <td>PENblue123</td><td>PEN</td><td>blue123</td></tr> </tbody> </table>	ItemName	Item	Code	PENblue123	PEN	blue123
ItemName	Item	Code						
PENblue123	PEN	blue123						

2. MERGE COLUMNS

- Combines two or more columns into a single column.
- Useful for creating keys, labels, addresses, full names, or joined values.

How to use :

Select Columns → Transform Tab → Merge Columns → Choose separator

MERGE COLUMN - OPTIONS

- Choose a separator (space, comma, dash, pipe "|", etc.)
- Choose "Without Separator" for direct join.
- You can create custom text and combine any number of columns.

Example :

First Name	Last Name	City	Merge Columns (Space)		Full Name	Location
Shivansh	Yadav	Lucknow			Shivansh Yadav	Lucknow
Rahul	Singh	Kanpur			Rahul Singh	Kanpur
Neha	Verma	Delhi			Neha Verma	Delhi

Use Cases



- ★ Create Full Name from First & Last.
- ★ Create Address from multiple parts.
- ★ Create Key columns.
- ★ Create Labels for reports & dashboards.

3. FILL DOWN

- Copies the value from the cell above into selected empty cells.
- Very useful for hierarchical or grouped data.

How to use :

Select Column → Transform Tab → Fill → Down

Region	Salesperson	Sales
North	Aman	12000
	Rohit	15000
	Neha	11000
South	Vivek	13000
	Karan	9000
	Priya	14000

Fill Down

Region	Salesperson	Sales
North	Aman	12000
North	Rohit	15000
North	Neha	11000
South	Vivek	13000
South	Karan	9000
South	Priya	14000

Use Cases

- ★ Fill missing Group/Category values.
- ★ Fill merged header values downwards.
- ★ Maintain consistency in reports.

4. FILL UP

- Copies the value from the cell below into selected empty cells.
- Less common but helpful in specific scenarios.

How to use :

Select Column → Transform Tab → Fill → Up

Date	Remarks
01-01-2024	
02-01-2024	
03-01-2024	Payment Received
04-01-2024	
05-01-2024	Order Shipped
06-01-2024	

Fill Up

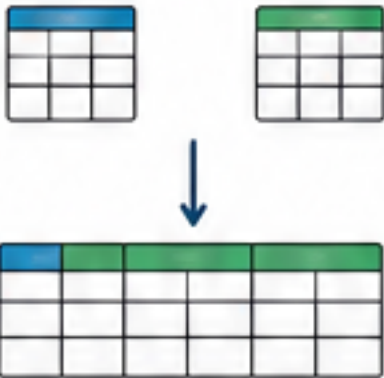
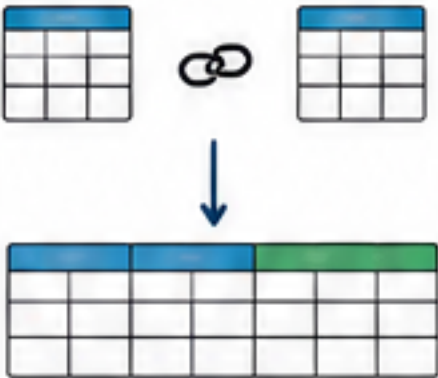
Date	Remarks
01-01-2024	Payment Received
02-01-2024	Payment Received
03-01-2024	Payment Received
04-01-2024	Order Shipped
05-01-2024	Order Shipped
06-01-2024	Order Shipped

Use Cases

- ★ Fill future dates with next known value.
- ★ Fill remarks or status upwards.
- ★ Handle planning data.

5. APPEND vs MERGE (JOIN)

Two important ways to combine data from multiple tables.

APPEND (UNION)	MERGE (JOIN)
- Stacks rows from two or more tables with the same structure. - Similar to UNION ALL in SQL. - Increases number of rows. 	- Combines columns from two tables based on a related (column/common key). - Similar to JOIN in SQL. - Increases number of columns. 

Join Types in Merge

- Left Outer (All from Left, Matching from Right)
- Right Outer (All from Right, Matching from Left)
- Full Outer (All from Both)
- Inner Join (Only Matching)
- Left Anti (Only Non Matching from Left)
- Right Anti (Only Non Matching from Right)



6. GROUP BY (AGGREGATION)

- Summarizes data using aggregation functions.
- Similar to Pivot Table but more flexible.

Example :

Region	Salesperson	Sales
North	Aman	12000
North	Rohit	15000
South	Vivek	13000
South	Karan	9000

Group By Region

Region	Total Sales
North	27000
South	22000

Use Cases

- ☆ Total Sales by Region
- ☆ Count by Category
- ☆ Average by Group
- ☆ Max/Min by Group
- ☆ Any custom summary

7. PIVOT / UNPIVOT

PIVOT COLUMN

- Turns unique values in a column into new columns.

Example :

Before (Long Format)

Month	Product	Sales
Jan	Pen	500
Jan	Book	800
Feb	Pen	600
Feb	Book	900

After (Wide Format)

Month	Pen	Book
Jan	500	800
Feb	600	900

UNPIVOT COLUMN

- Turns multiple columns into attribute-value pairs.

Example :

Before (Wide Format)

Month	Pen	Book
Jan	500	800
Feb	600	900

After (Long Format)

Month	Attribute	Value
Jan	Pen	500
Jan	Book	800
Feb	Pen	600
Feb	Book	900

Use Cases

- ☆ Reshape data for reporting & dashboards.
- ☆ Convert wide data to long for analysis.
- ☆ Make data easy to plot in charts.

8. CONDITIONAL & CUSTOM COLUMNS

A. CONDITIONAL COLUMN

- Creates a new column based on a condition.
- Uses if-then-else logic.

Example :

Sales	Category
12000	High
8000	Medium
3000	Low

Condition Example

If Sales > 10000 → "High"

If Sales between 5000-10000 → "Medium"

Else → "Low"

B. CUSTOM COLUMN

- Create a new column using a custom formula.
- Uses Power Query formula language (M).

Example : Add 18% GST Amount

Amount	GST Amount (18%)
1000	180
2500	450
5000	900

Formula (M Code)

= [Amount] * 0.18

9. FOLDER IMPORT & COMBINING MONTHLY FILES

A. IMPORT DATA FROM FOLDER

- Import multiple files from a folder.
- Power Query reads all files and combines them.



B. COMBINING MONTHLY FILES

- Very useful for reports which receive files every month.
- Steps: Get Data → From Folder → Transform Sample File → Combine → Clean → Load

Use Cases

- ☆ Monthly Sales Reports
- ☆ Bank Statements
- ☆ Inventory Reports
- ☆ MIS Reports
- ☆ Auto Combine New Files

10. PARAMETERS (BASIC)

- Parameters store values that can be reused in queries.
- Types: Text, Number, Date, True/False.
- Used for dynamic queries and better maintainability.

Example :

Parameter Name	Value	Usage in Query
StartDate	01-04-2024	Filter rows where [Order Date] >= StartDate



11. BASIC M AWARENESS

- Power Query uses M Language behind the scenes
- Every step you apply is converted to M Code.
- View M Code: Home Tab → Advanced Editor
- Useful for advanced users and custom transformations.

Example M Code

```
let
    Source = Excel.Workbook(File.Contents("C:\Sales.xlsx")),
    Sheet1_Sheet = Source[[Item="Sheet1",Kind="Sheet"]][Data],
    Changed_Type = Table.TransformColumnTypes(Sheet1_Sheet,
        {"Sales", type number})
in
    Changed_Type
```

12. PRACTICAL WORKFLOW (SUMMARY)

1	1 Get Data	Import data from any source.	
2	2 Transform	Clean, standardize & shape your data.	
3	3 Combine	Append / Merge / Group / Pivot as needed.	
4	4 Enrich	Add conditional columns, custom columns.	
5	5 Validate	Check data quality (errors, blanks, types).	
6	6 Load	Load cleaned data to Excel / Data Model.	

13. PARAMETERS & BASIC M AWARENESS



A. PARAMETERS



Parameters store values that can be used multiple times in queries.



Change the value once, it updates everywhere it is used.



Types : Text, Number, Date, True/False, Any

Why Use Parameters?

- ✓ Centralized control of values
- ✓ Reduces errors
- ✓ Makes queries dynamic & reusable
- ✓ Great for reports with filters like date, region, etc.

Example : Parameter for Start Date

Parameter Name	StartDate
Type	Date
Current Value	01-04-2024
Used In Query	Filter rows where Date >= StartDate



Real Life Use Case

If you create a parameter for Start Date, you can change the date once and all reports will update automatically.

How to Create a Parameter in Power Query

1 Home Tab



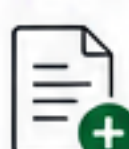
Go to Home Tab

2 Manage Parameters



Click on Manage Parameters

3 New Parameter



Select New Parameter

4 Set Details



Give Name, Type, Current Value

5 Use in Query



Use the Parameter in your query

B. BASIC M AWARENESS



Power Query uses M Language behind the scenes.



M is case-sensitive.



Every step you perform is recorded in M Code.



You can view and edit M Code for advanced control.

Example : Simple M Code

```
let
    Source = Excel.Workbook(File.Contents("C:\Sales.xlsx"), null, true),
    Sheet1_Sheet = Source[Item="Sheet1",Kind="Sheet"][Data],
    Promoted_Headers = Table.PromoteHeaders(Sheet1_Sheet,
        [PromoteAllScalars=true]),
    Changed_Type = Table.TransformColumnTypes(Promoted_Headers,
        {"Sales", type number}, {"Date", type date})
in
    Changed_Type
```

What does this code do?



1. Source

Reads Excel File



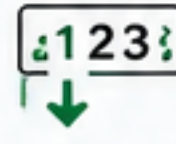
2. Sheet1_Sheet

Selects Sheet1 data



3. Promoted_Headers

Promotes first row as headers



4. Changed_Type

Changes column data types



5. Output

Returns the final transformed table

Key M Functions You Already Saw

Excel.Workbook()

Imports data from Excel file.

Table.PromoteHeaders()

Converts first row into column headers.

Table.TransformColumnTypes()

Changes column data types.

Table.SelectRows()

Filters rows based on conditions.

Table.RemoveColumns()

Removes selected columns.

Table.Group()

Groups data & applies aggregation.

Note

You don't need to write M code from scratch. Just understand that every action = a step in M.



15. BEST PRACTICES

- ✓ Rename queries meaningfully.
- ✓ Remove unnecessary steps.
- ✓ Change data types early in the process.
- ✓ Remove duplicates and blanks before analysis.
- ✓ Use Parameters for flexibility and reusability.
- ✓ Always check data quality before loading.
- ✓ Keep steps simple, consistent and reusable.
- ✓ Document your workflow with clear steps.



Pro Tip : Build a workflow that others can reuse with zero confusion.

16. REAL LIFE EXAMPLES (END TO END SCENARIOS)



RAW DATA
(Multiple Files)



POWER QUERY



CLEAN DATA



REPORT / DASHBOARD

Scenario	What We Do in Power Query	Business Impact
Monthly Sales Report	Import monthly files, combine, clean, transform and create summary.	Saves time, faster reporting, consistent data.
Bank Statements	Clean transactions, remove blanks, standardize categories and filter.	Accurate financial analysis, fraud detection.
Inventory Report	Combine multiple files, fix data types, remove errors and create KPIs.	Better inventory tracking, reduced stock issues.
HR Analytics Report	Clean employee data, standardize departments, group and summarize.	Better HR insights, attrition & performance tracking.
Marketing Campaign Data	Append campaign files, clean metrics, standardize channel names.	Better campaign analysis, clear ROI insights.

17. PRACTICAL BUSINESS USE CASES

1

Sales Consolidation

Combine multiple branch sales files and create a single consolidated report.



2

Expense Reconciliation

Clean expense data, remove duplicates, fix categories and match with budgets.



3

Customer Analysis

Clean customer master, standardize city/state names and remove invalid records.



4

Financial Reporting

Transform raw GL data, group by accounts and create Profit & Loss / Balance Sheet.



5

Operational Dashboard

Prepare clean, structured data for dashboards in Excel / Power BI.



18. INTERVIEW QUESTIONS WITH ANSWERS

Q. No.	Interview Question	Expected Answer (Summary)
1	What is the difference between Append and Merge in Power Query?	→ Append (Union) stacks rows from tables. Merge (Join) combines columns based on a key.
2	What are the types of Joins supported in Power Query?	→ Left Outer, Right Outer, Full Outer, Inner Join, Left Anti, Right Anti.
3	What is Group By in Power Query?	→ Group By aggregates data using functions like Sum, Count, Avg, Min, Max etc.
4	What is the difference between Pivot and Unpivot?	→ Pivot converts row values into columns. Unpivot converts columns into attribute-value pairs.
5	When would you use a Conditional Column?	→ When you want to create a new column based on a condition using if-then-else logic.
6	What is the use of Custom Column?	→ To create a new column using a custom formula or M code.
7	How do you import multiple files from a folder?	→ Get Data → From Folder → Combine & Transform or Combine Files.
8	How do you combine monthly files?	→ Import files → Combine → Transform Sample File → Clean → Load.
9	What are Parameters in Power Query?	→ Parameters store values and make queries dynamic and reusable.
10.	Why is basic M code knowledge important?	→ It helps in understanding what happens behind the scenes and allows advanced customizations.

19. MORE IMPORTANT CONCEPTS TO REMEMBER

DATA QUALITY FIRST



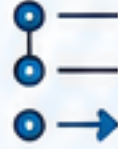
Always check for Errors, Blanks, Duplicates & Incorrect Data Types before analysis.

SOURCE IS KING



Good source data + Good transformations = Accurate insights.

APPLIED STEPS ARE POWERFUL



Every step is recorded. You can go back, modify or remove any step.

REFRESH IS MAGIC



Any change in source ? Just click Refresh All and get updated results.

LOAD SMARTLY



Load to Table, Pivot, Data Model as per your reporting requirements.

20. COMMON MISTAKES TO AVOID

- ✗ Changing data types at the end.
- ✗ Not removing duplicates or blanks early.
- ✗ Using too many unnecessary steps.
- ✗ Not naming queries properly.
- ✗ Ignoring data quality checks.
- ✗ Loading huge dirty data directly.
- ✗ Not using Parameters for reusability.
- ✗ Not refreshing queries regularly.

21. POWER QUERY WORKFLOW (SUMMARY)

1. Get Data



Import data from any source.



2. Transform



Clean, shape, filter and standardize.



3. Combine



Append, Merge, Join as needed.



4. Enrich



Add columns, conditions, custom logic.



5. Validate



Check data quality, errors, blanks.



6. Load



Load to Excel / Data Model / Power BI.



Power Query is not just a tool, it is a complete Data Preparation Engine. Master it and you will save time, reduce errors and improve the quality of your analysis.

Clean Data
Better Insights
Better Decisions

